جامعة أم القرى
UMM AL-QURA UNIVERSITY

College of Computing

Cybersecurity Department

RESEARCH PROJECT

Comprehensive Survey on Alexa Forensics: Data Storage and Artifact Retrieval on Alexa versus Alexa

Submitted by:

Ghadah [redacted] [redacted]
Reemas [redacted] [redacted]
Kholod [redacted] [redacted]
Sarah Al-Sulami [redacted]

Section Number: 2

Instructor: Dr. Daniah [redacted]

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Abstract

Amazon Alexa and Echo devices are now widely used in homes and smart environments, which raises important questions about how the system stores user interactions and device events. Although Alexa is designed to offer simple voice control and smart-home automation, its architecture spreads data across three main areas: the Echo device, the Alexa mobile application, and Amazon's cloud services. This review explores where Alexa keeps these data and what forensic tools can recover them. Previous studies show that valuable artifacts can be retrieved through mobile device analysis, cloud API access, and hardware methods such as UART, JTAG, and chip-off extraction. The findings suggest that, even with recent security improvements, a multi-layered forensic approach can still successfully reconstruct user activities and system behavior within the Alexa ecosystem.

Keywords: Amazon Alexa, IoT forensics, Digital forensics, Alexa artifacts

1. Introduction

Virtual assistant-based systems like Amazon Echo and Google Home gained increasing popularity over the last few years, and are expected to become more important in the future. These systems are used in daily life to control smart home environments, order products, issue queries or organize someone's life [6]. Amazon Alexa has become a central component of modern smart-home environments, with more than 100 million devices deployed globally [14]. Through voice interaction, automation features, and integration with a wide range of third-party skills, Alexa devices generate continuous streams of data related to user activity, device behavior, and cloud-based processing. This creates a distributed data ecosystem that spans local device storage, mobile companion applications, cloud services, and external service providers.

As Alexa ecosystems continue to expand, questions related to security, privacy, and digital evidence have gained increasing attention. Prior studies have shown that Alexa's reliance on cloud processing limits the amount of information stored locally on devices, while access to cloud data may require authorization procedures that affect the timeliness of investigations. At the same time, the involvement of third-party skills introduces additional uncertainty, as data-handling practices and review mechanisms vary and may allow unintended behavior or misuse [15].

One example illustrating these concerns is the Alexa vs Alexa (AvA) attack. In this attack, an adversary plays audio files containing voice commands through the device's own speaker via Bluetooth or a malicious media stream causing Alexa to interpret them as legitimate user commands [5]. This self-issued command capability allows unauthorized actions to be executed without a human speaker present, raising concerns about both device security and the reliability of recorded evidence.

Given the growing role of voice-controlled systems in everyday settings, understanding how Alexa stores user interactions and evaluating the forensic methods used to retrieve these records have become important. This study aims to provide a comprehensive review of the literature published between 2020 and 2025 concerning Alexa data storage and forensic artifact acquisition. The objectives are to (i) analyze where and how Alexa stores user interactions and device events, and (ii) evaluate the effectiveness of current forensic tools and acquisition techniques documented in prior research.

2. Literature Review

2.1 Overview of Alexa Architecture and Data Ecosystem

The Amazon Alexa system consists of both physical and cloud-based components that work together to enable voice interaction, smart home control, and data management. The architecture can be broadly divided into two main domains: the user environment and the cloud backend.

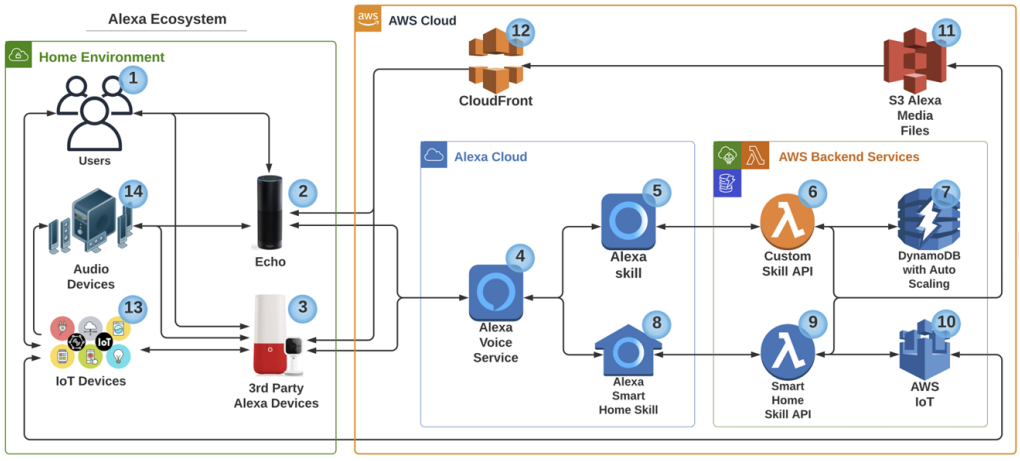


Figure 2.1: Overview of the Alexa Ecosystem [10].

In the user environment, individuals interact with the system through voice commands or connected devices such as Echo devices. Echos are Alexa-powered smart speakers from Amazon [8], which continuously listen for the wake word "Alexa," record user commands, transmit them to the cloud for processing, and then deliver the response audibly. Third-party devices, IoT devices, and audio peripherals further extend the interaction capabilities, allowing the execution of user commands in real-world settings.

On the cloud side, the Alexa Voice Service (AVS) provides automatic speech recognition (ASR), natural language understanding (NLU), and text-to-speech (TTS) functionalities to interpret user intent and determine the appropriate skill to execute. Voice skills, whether pre-installed or custom-developed using AWS Lambda, perform the requested

operations. Skill-related data are stored in DynamoDB, smart devices are managed via AWS IoT, and static content such as audio and images is stored in Amazon S3 and distributed efficiently using CloudFront. All components operate in coordination to ensure that user commands are captured locally, interpreted in the cloud, executed through skills or device integration, and returned to the user in real-time [11].

2.2 Alexa Data Storage Locations

Understanding how and where Alexa stores user interactions and device events is foundational to digital forensic investigations. Prior research demonstrates that the Alexa ecosystem distributes its data artifacts across three primary layers the cloud service, the mobile companion application, and the local device storage embedded within Echo hardware as shown in Figure2.2. Each of these layers contains distinct forensic traces, and together they form a multilayered evidence structure that investigators can analyze to reconstruct user activity, device behavior, and potential security incidents. This aligns with the multi-source forensic model described in the Echo Show forensic analysis, which emphasizes that meaningful forensic evidence emerges only when artifacts from hardware, client devices, and the Alexa cloud are examined collectively[19].

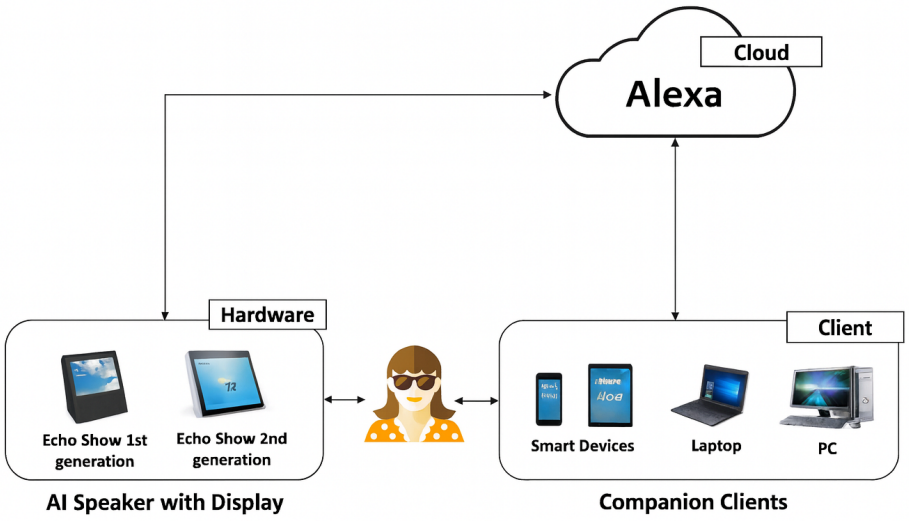


Figure 2.2: Amazon Echo Show Ecosystem[19]

Furthermore, network-level analyses of Alexa communications show that even when payloads are encrypted, metadata exchanged between Echo devices and Amazon servers can reveal operational behaviors, command execution patterns, and third-party service interactions[2]. Also, user studies reveal that data may be sent to third-party servers when users interact with non-Amazon skills, meaning the forensic evidence trail extends beyond Amazon's infrastructure[16].

2.2.1 Cloud Storage (Primary Forensic Source)

The Alexa cloud is consistently identified as the most comprehensive and authoritative repository of user interactions within the Amazon ecosystem. Research focusing on cloud centric forensics demonstrates that when users interact with Alexa-enabled devices, the resulting artifacts including voice recordings, transcribed commands, timestamps, skill invocation logs, shopping list entries, to-do lists, multimedia records, and device metadata are stored remotely within Amazon's servers[19].These cloud-native artifacts can include behavioral data such as search queries, smart home commands, multimedia requests, weather inquiries, and prior user conversations, all of which can be retrieved through authenticated API access or forensic tools such as the Cloud-based IoT Forensic Toolkit (CIFT), as documented by previous Echo forensic studies[19].Alexa cloud storage also maintains historical user behavior logs, including photos and videos taken on Echo Show devices, which are uploaded automatically to the cloud and cannot be fully reconstructed solely from the device's local memory. The Echo Show forensic case study demonstrates this clearly: while the device stored timestamps and hash values of captured media, the actual images were located only in the Alexa cloud account, enabling investigators to reconstruct events and infer user interests and activity patterns[19]. Additionally, the cloud logs all metadata associated with network communication, as evidenced by the Alexa Traffic Traces dataset. While payloads are TLS-encrypted, investigators can observe the full list of Amazon endpoints contacted during interactions providing indirect evidence of command issuance and system activity[2]. Given the completeness, accuracy, and permanence of cloud-stored data, the cloud operates as the primary forensic source, serving as the central data repository in the Alexa ecosystem and containing the most complete representation of user interactions.

2.2.2 Companion App Storage (Android/iOS)

The Alexa companion application for Android and iOS maintains a significant amount of locally stored data that supplements cloud-native artifacts. The Echo Show forensic analysis identifies several locally stored files such as service.identity.xml, Web Data, RKStorage, and DataStore.db—that collectively store account identifiers, cached Alexa responses, skill usage history, alarm/timer data, browsing activity, and images used in Alexa replies[19]. Importantly, the companion app stores CardID interaction metadata, which links local app logs to cloud-stored history, enabling investigators to correlate events across layers. The details are shown in the Table 2.1.

Table 2.1: Companion App Forensic Artifacts (Android/iOS)

Artifact Type	**What Stores**
Account metadata	Amazon account name, email, identifiers
CardID interaction logs	Links app logs to cloud history, used to correlate events
Cached system replies	Images, text replies, alarm/timer updates synchronized from cloud
Browser history	URLs accessed through Echo Show browser
OAuth tokens	Tokens may allow retrieval of cloud logs programmatically
Skill usage metadata	Local logs showing installed and activated skills, timestamps

2.2.3 Echo Device Local Storage (Limited, but Valuable)

Echo devices maintain limited local storage, but the artifacts they contain have substantial forensic significance. The Echo Show forensic disassembly reveals that internal storage uses an Ext4 filesystem consisting of multiple partitions, with forensically relevant data residing primarily in the android data partition[19]. So, The Table2.2 shows the local storage on Echo devices.

Table 2.2: Echo Device Local Storage Artifacts and Their Forensic Value

Artifact Type	**What the Device Stores**
System logs	Boot logs, camera activation events, LED state changes, and wake-word detections that record device behavior.
Silk browser data	Stored email credentials, browsing history, and cached sessions generated when using the Echo Show browser.
Multimedia metadata	Hash values, timestamps, and filenames for photos and videos captured or processed by the device.
Cached video fragments	Partial cached chunks of streamed video (e.g., YouTube, Dailymotion) that can be reconstructed into playable media.

Continued on next page

Artifact Type	What the Device Stores
False wake-word logs	Records of unintended wake-word activations triggered without user action, useful for timeline reconstruction.
LED state logs	Records of the light-ring indicators and LED behaviors associated with system messages and skill interactions.

2.3 Alexa Vulnerabilities and the Alexa vs Alexa (AvA) Attack

Amazon Alexa operates through an integrated architecture that spans device-level audio processing, cloud-based interpretation, and skill execution. While this ecosystem is designed to provide seamless interaction and secure user experiences, multiple studies have shown that several components remain vulnerable to exploitation. These weaknesses can enable adversaries to influence device actions, mislead users, or obtain access to information that should remain private.

This section outlines the vulnerabilities identified across the Alexa ecosystem, highlighting weaknesses in voice capture, communication channels, and skill interactions. It then introduces the Alexa vs Alexa (AvA) attack, a notable example demonstrating how an Echo device can be manipulated using audio generated by the device itself effectively turning Alexa into both the attacker and the victim.

2.3.1 Alexa Vulnerabilities

Previous research has shown that security vulnerabilities can emerge at various stages of the Alexa processing pipeline, ranging from the moment a voice command is captured to the cloud infrastructure where interpretation and skill logic occur.

The study reviews a number of documented attacks on Alexa systems, either tested or reported in the literature. These attacks include Skill Squatting, which experiments demonstrated to have a success rate of 92.6% when testing 27 pairs of similar skills, as well as the targeted variant Spear Skill Squatting, which uses localized or categorical names to increase deception rates. Other attacks include Voice Masquerading, where malicious skills were able to switch sessions or simulate closure while continuing to record the user by bypassing the voice response mechanism.

The study also discusses ASR Adversarial attacks, which allow execution of hidden voice commands that the user cannot perceive, and Voice Capture attacks, in which Echo devices recorded human and television sounds without activation by the wake word during a 21 day experiment. Additionally, ultrasonic-based attacks such as the Dolphin Attack

were successful at distances of up to 1.65 meters, and Man-in-the-Middle Voice Hijacking attacks enabled interception and modification of commands in real time. Finally, Traffic Fingerprinting attacks were described, allowing the inference of user commands by analyzing encrypted network traffic.

These findings highlight multiple points of vulnerability and provide a foundation for understanding the associated threats and the security measures needed [7].

Table 2.3 summarizes the primary categories of vulnerabilities reported in the literature.

Table 2.3: Summary of Alexa Attack Surfaces and Associated Attacks [14]

Attack Surface	**Attack Name**	**Description**
Voice Capturing	Remote Voice Attack	Occurs when attackers broadcast Alexa wake words and commands through external speakers. Since no voice authentication is enforced, the Echo device responds to any audible command.
	Dolphin Attack	Uses voice commands modulated onto ultrasonic frequencies, inaudible to humans but detected by microphones, allowing activation from short distances.
	Man-in-the-Middle (Voice Hijacking)	An IoT device intercepts the user's interaction with Alexa by jamming commands, recording them, modifying requests, and returning falsified responses.
Voice Transmission	Traffic Fingerprinting Attack	Encrypted traffic between Alexa devices and cloud servers is monitored to infer user commands based on packet size, timing, and patterns.
Alexa Voice Service (ASR/NLU/TTS)	Skill Squatting	Malicious skills are created with invocation names that sound similar to legitimate ones, exploiting speech recognition errors and pronunciation variations [13].
Alexa Voice Skill	Malicious Skills Publication	Attackers upload skills that intentionally violate policies, such as collecting user data. Weaknesses in the review process allow these skills to pass.
	Masquerading Attack	A malicious skill imitates legitimate skills or Alexa itself to trick users into providing sensitive information.
	Silent Eavesdropping Attack (Checkmarx Attack) [7]	A malicious skill suppresses Alexa's readback, stays active silently, and records user speech without detection.
Lambda Functions	Command / SQL Injection via Voice Input	Poorly validated Lambda functions interpret crafted voice inputs as commands, enabling SQL injection or backend execution.
Amazon S3 Buckets	S3 Bucket Misconfiguration	Misconfigured S3 buckets expose assets; modified files can cause Alexa to deliver manipulated responses.

2.3.2 The Alexa vs Alexa (AvA) Attack

The Alexa vs Alexa (AvA) attack leverages a core weakness in Amazon Echo devices: the inability to distinguish between the user's real voice and audio played through the device's own speaker. By preparing audio clips that include the wake word followed by commands whether generated through TTS, adversarial audio, or human recordings an attacker can trigger the device to issue and execute commands on itself. These clips are delivered either through Bluetooth injection, where the attacker pairs with the device and streams the audio directly, or through a malicious radio skill that embeds hidden wake

words and commands within its audio stream. Once played, the device's microphones capture the audio from its own speaker, interpret it as a legitimate user request, and execute it normally, forming the core of the self activation vulnerability. After gaining command execution, the attacker can enable long term surveillance through a malicious skill that records user utterances and relays them to an external server, which in turn forwards modified requests to Amazon and returns processed responses allowing the attacker to mediate and manipulate all interactions. Techniques such as SSML-based delays help maintain persistent sessions, while self-issued commands can terminate or resume control at will. This enables unauthorized actions including smart home manipulation, unauthorized purchases, data theft, eavesdropping, and impersonation of Alexa all without breaking network security, as the attack relies entirely on exploiting the device's own speaker against itself [5].

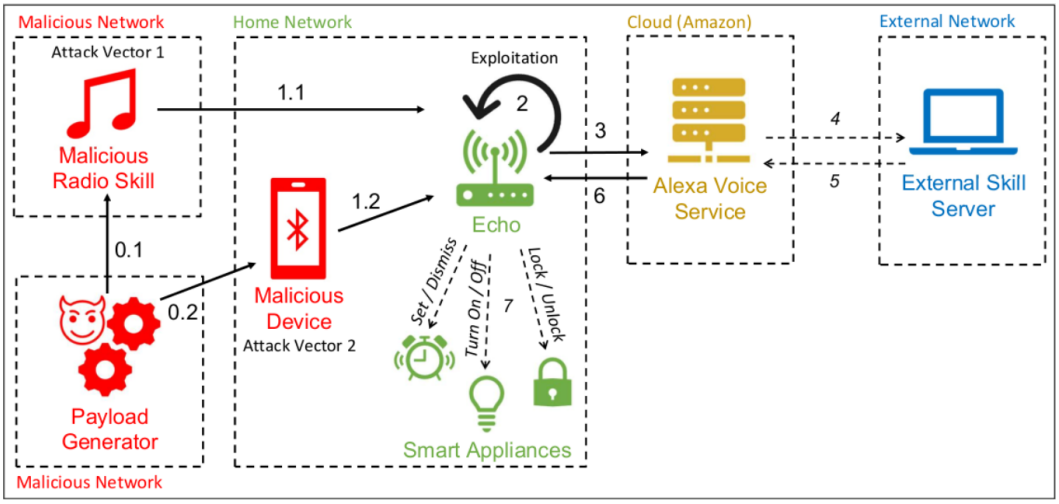


Figure 2.3: Alexa vs Alexa (AvA) attack workflow illustrating self-issued voice commands via Bluetooth injection and malicious radio skills [5].

Table 2.4: Summary of AvA Attack Components and Mechanisms [5]

Category	Component	Description
Attack Vectors	Bluetooth Injection	The attacker connects via Bluetooth and streams audio containing wake-word commands, which the Echo device executes immediately.
	Radio-Skill Injection	A malicious radio skill broadcasts audio with embedded commands; when the victim plays the stream, the Echo device activates and executes those commands.
Payload Types	TTS Commands	Commands generated using text-to-speech; the clearest and most reliable payload type.
	Adversarial Audio	Audio crafted with subtle perturbations that fool ASR systems while sounding normal to humans.
	Human Recordings	Pre-recorded human voice commands; functional but less accurate and more traceable.
Core Vulnerabilities	Self-Issue Vulnerability	The Echo device can hear and execute commands coming from its own speaker output.
	Full Volume Vulnerability	During Bluetooth streaming, Echo remains at maximum volume after self-issued commands, making injected audio easier for microphones to capture.
Post-Exploitation	Mask Attack Skill	A malicious skill that stays open, listens silently, collects sensitive data, and impersonates Alexa responses.
Persistence	SSML Break Chain	By chaining hundreds of SSML `<break>` tags, a malicious skill stays active for long periods, enabling persistent eavesdropping.

2.4 Forensic Tools and Acquisition Methods for Alexa Artifacts

Forensic acquisition of Alexa artifacts requires a multi layered strategy that integrates local device imaging, mobile application examination, and cloud data retrieval. Previous Alexa focused research, such as the work by [1], used FTK Imager to acquire application data including logs, configuration files, and interaction records from mobile devices running the Alexa companion app, while Autopsy was applied in the post acquisition phase to analyze these artifacts and compare them with cloud stored metadata [1]. Comprehensive IoT forensic research focuses on an integrated approach that combines network data, application storage, and cloud information to fully reconstruct user actions and system events[3]. Smart home investigations often rely on disk imaging, network packet capture, and artifact extraction to support digital investigations [11]. Additionally, companion apps, APIs, and cloud interfaces serve as key sources of behavioral evidence, enabling reconstruction of device interactions, user commands, and environmental events [10].

2.4.1 Cloud Acquisition Techniques

Cloud acquisition techniques focus on retrieving remotely stored Alexa artifacts from Amazon's cloud infrastructure using authenticated APIs, legal procedures, and network-based evidence correlation. Cloud services represent the central storage layer for Alexa artifacts, including timestamps, metadata, and voice recordings. Alexa devices automatically upload voice interactions to Amazon servers, sometimes including unintended recordings, creating both valuable forensic evidence and privacy risks [1]. Table 2.5 illustrates examples of voice recordings retrieved from the Alexa cloud account, showing both intentional and unintentional interactions alongside their activation methods and privacy implications. Notably, unintended background conversations were stored remotely without explicit user consent, highlighting privacy weaknesses in cloud based retention.

Table 2.5: Alexa Voice History from Cloud Account [1]

Recording ID	**Timestamp**	**Recorded Content**	**Activation Method**	**Privacy Status**
RH001	2024-11-03 10:00:00	"What's the weather?"	Voice wake word	Expected recording
RH002	2024-11-03 10:10:00	"Play music"	Voice wake word	Expected recording
RH003	2024-11-03 10:15:00	Background conversation	None (no activation)	Privacy concern noted
RH004	2024-11-03 10:20:00	"Settings access"	Tap	Expected recording

IoT forensic literature identifies cloud data as a "central hub," requiring API-based extraction, legal requests, and correlation with device identifiers to contextualize stored artifacts [3]. Research on Echo devices reinforces the cloud's importance by showing that Automatic Speech Recognition and processing occur remotely and that most voice data is not retained locally, making cloud acquisition essential [18]. As illustrated in Figure 2.4, speech recognition errors where Alexa heard 'El examen' as the wake word. This shows that Alexa depends on cloud processing, so cloud data is needed for forensic investigations.

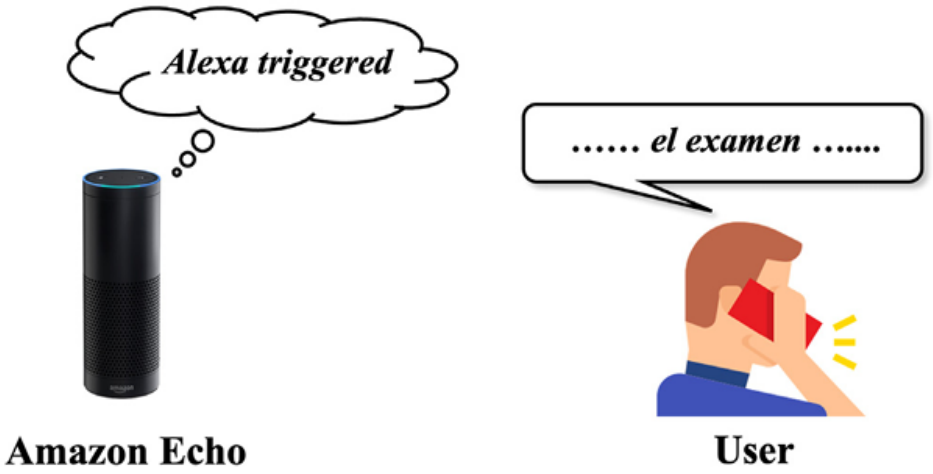


Figure 2.4: Speech recognition error: "El examen" interpreted as Alexa triggers the device[18].

IoT forensic research consistently highlights that cloud acquisition is essential when investigating smart assistants such as Amazon Alexa, since most operational evidence is stored remotely rather than on the physical device. Network-based acquisition supports this process by capturing Wi-Fi traffic between the device and cloud servers, allowing investigators to identify API endpoints, request patterns, timestamps, and metadata transmitted by the device. These packet traces help infer which artifacts exist in the cloud and guide targeted retrieval through authenticated API requests or legal disclosure procedures. While invasive methods such as vulnerability exploitation, UART/JTAG access, or chip-off extraction are often used in other IoT investigations, they are generally ineffective for Alexa due to encrypted storage and locked hardware. As a result, cloud-centric acquisition supported by network packet analysis and API-level extraction represents the most reliable and effective method for obtaining forensic artifacts from Alexa environments [11].

Cloud acquisition is further reinforced by research showing that Alexa's web platform communicates with Amazon's cloud using a large collection of undocumented APIs that return detailed forensic artifacts[12]. By intercepting HTTPS traffic from the Alexa web interface, investigators were able to bypass TLS encryption and HSTS protections to identify dozens of API endpoints responsible for delivering historical activities, cards, utterances, device metadata, household information, and raw voice transcripts. This method revealed that nearly all meaningful Alexa evidence such as timestamps, interaction logs, device identifiers, and user-associated metadata is stored exclusively in the cloud, rather than on the Echo device itself. The study also found that user deletion does not consistently remove all data; voice recordings may be deleted while metadata persists, or activity entries remain visible through API calls even after being cleared from

the dashboard [12]. These results confirm that acquisition through cloud APIs is not only feasible but often exposes residual evidence that survives user-initiated deletion, making it a crucial forensic approach for smart speaker investigations.

2.4.2 Mobile Device Acquisition Tools

Once the device is located, the primary objective is to extract useful data that can guide the investigation toward a clear conclusion. In this context, mobile devices associated with Alexa serve as an essential forensic source because the companion application functions as a secondary repository for configuration data, logs, and interaction history. These devices often retain evidence that is not available directly from the Alexa hardware. Mobile devices linked to Alexa retain extensive forensically relevant data because companion applications record configuration files, logs, and voice interaction history. Deleted commands can remain recoverable for 16–32 days, demonstrating persistent local storage and incomplete deletion mechanisms [1]. Smart home forensic studies further confirm that companion apps store caches, databases, images, events, and device listings, all of which can be extracted through logical acquisition tools such as Oxygen Forensic Detective, FTK Imager, and Autopsy, or through direct inspection of application storage [10]. Such artifacts enable investigators to reconstruct detailed behavioral patterns including motion events, user interactions, and device state changes supporting high resolution timeline reconstruction [11].

In addition to application level artifacts, recent forensic examinations of Amazon Echo devices highlight the role of device level acquisition tools. For example, Crasselt and Pugliese successfully extracted data from an Amazon Echo Show 15 by identifying undocumented *eMMC test points* on the device's mainboard and performing a full *eMMC memory dump*, enabling the recovery of system partitions and locally stored logs [4]. This demonstrates that low level hardware acquisition similar in concept to JTAG or chip-off techniques can complement mobile based acquisition when cloud linked devices store critical evidence internally.

Therefore, smartphone based extraction combined with device level acquisition techniques represents a critical forensic pathway, as it integrates metadata, user content, configuration parameters, cloud interaction traces, and cross device activity into a comprehensive evidence set [3].

2.4.3 Echo Device Hardware and Firmware Forensics

Echo devices retain system files, logs, and configuration data locally, making them potential sources of forensic evidence accessible through hardware-level extraction techniques. Some studies have shown that certain Echo models provide exposed debug interfaces, which can enable JTAG or chip-off extraction, root shell access, and SD-boot attacks.

These methods allow investigators to retrieve firmware, system logs, encryption keys, and authentication tokens [18]. Studies in smart home forensics highlight that modern IoT devices often require a combination of logical and physical acquisition methods, including UART exploitation, NAND extraction, and file system analysis, to obtain system-level artifacts, especially when cloud or remote data is incomplete [11].

2.5 IoT Forensics Challenges in Smart Speaker Ecosystems

Smart assistants such as Amazon Alexa or similar AI-based voice assistants operate within complex IoT environments that introduce multiple forensic challenges. These devices continuously interact with cloud services, companion mobile applications, and interconnected home automation systems, which complicates both data acquisition and analysis [9].

One major issue is the heterogeneous nature of smart speaker architectures. IoT devices often vary in connection protocols, data formats, and software architectures, making it difficult for investigators to access and analyze data consistently across devices [9]. This lack of uniformity is especially significant for smart speakers, where logs, configuration files, and interaction data may be stored using undocumented or inaccessible formats.

Smart speakers also face challenges related to limited local storage and logging capabilities. IoT devices generally have restricted storage and processing power, which can lead to overwritten logs, fragmented data, and the loss of crucial evidence [9]. Because smart speakers rely heavily on cloud processing, they typically store only minimal temporary data locally, making on-device forensic acquisition incomplete or insufficient.

Cloud dependency further complicates investigations. Most smart speaker interactions such as voice queries, device events, and behavioral patterns are sent to remote servers. IoT ecosystems frequently rely on cloud infrastructure, which introduces problems such as limited investigator access, authorization barriers, and the possibility of remote data modification or deletion [9]. These challenges apply directly to smart speakers, where essential evidence is often accessible only through cloud accounts and vendor-controlled APIs.

Another challenge arises from the deployment of smart speakers in uncontrolled environments. IoT devices are often used in unregulated settings, increasing the likelihood of evidence tampering, altered logs, or missing data [9]. Because smart speakers are typically located in homes and offices, any user can reset the device, unlink accounts, or modify stored configurations, resulting in lost or contaminated evidence.

The lack of standardization across smart speaker vendors also affects forensic investigations. IoT systems suffer from limited standardization, which complicates tool development and data extraction processes [9]. Each vendor may implement different logging

mechanisms, access controls, cloud interfaces, and security models. This inconsistency makes it difficult to establish unified forensic procedures for smart speaker ecosystems.

Smart speakers are also highly interconnected devices. IoT forensics requires examining device data, communications between devices, and interactions with supporting systems [9]. For smart speakers, this means investigators often need to gather evidence not only from the speaker itself but also from routers, smartphones, companion applications, and cloud dashboards. This interconnectedness expands the scope of forensic analysis and complicates chain of custody management.

IoT devices are increasingly targeted by sophisticated malware, which can impact the integrity of stored data and evidence availability [9]. In smart speaker ecosystems, these threats may include malicious skills, unauthorized audio commands, and device hijacking, all of which are challenging to trace due to limited logging and distributed data storage across local and cloud environments.

3. Methodology

3.1 Research Question

This research investigates the forensic potential of Amazon Alexa by examining both its data storage architecture and the available methods for artifact acquisition. The study is guided by two primary research questions:

1. **Where and how does Alexa store user interactions and device events?**
2. **Which forensic tools and acquisition methods exist for retrieving Alexa artifacts?**

These questions aim to provide a comprehensive understanding of Alexa's internal data handling and the forensic approaches currently available. Answering them will help identify gaps in existing methods and suggest areas for future research.

3.2 Data Collection

This study adopts a structured literature-based data collection approach to examine Alexa forensic artifacts and acquisition methods. Since the research focuses on previously published work rather than empirical experimentation, data were collected exclusively from scholarly databases and reputable digital libraries. The following academic databases were used to retrieve relevant studies:

- IEEE Xplore: for peer-reviewed articles on IoT forensics, cloud acquisition, and Alexa security mechanisms.
- ScienceDirect: for studies on digital forensics, smart-speaker ecosystems, and encrypted network analysis.
- Google Scholar: to capture additional grey literature, preprints, and conference papers not indexed elsewhere.

To ensure comprehensive coverage, multiple search keywords were used, including: "Alexa forensics," "Echo device analysis," "IoT forensic acquisition," "smart speaker vulnerabilities," "cloud forensic techniques," "Alexa artifacts," "Alexa vs Alexa attack," and "smart home privacy." The search period was restricted to **2020–2025**, aligning with the timeline of modern Alexa architectures and recent forensic techniques.

3.3 Selection Criteria

To maintain academic rigor and ensure relevance to the research questions, explicit inclusion and exclusion criteria were applied.

Inclusion Criteria

- Studies published between 2020 and 2025.
- Research explicitly focused on Alexa, Echo devices, IoT forensics, smart speaker security, or forensic acquisition methods.
- Papers that examined cloud artifacts, mobile app storage, hardware-level forensics, or Alexa vulnerabilities.
- Peer-reviewed journals, conferences, and reputable technical reports.

Exclusion Criteria

- Publications unrelated to digital forensics or focusing on non-Alexa IoT devices unless directly comparable.
- Papers lacking methodological clarity or technical depth.
- Duplicate studies, outdated versions, or non-scholarly blogs and vendor marketing materials.

After applying these criteria, the final dataset consisted of the 20 references shown in the end of our project's document.

3.4 Data Analysis

A qualitative comparative analysis approach was used to evaluate and synthesize findings across the selected studies. Each paper was examined for:

1. What data artifacts were retrieved (cloud, mobile app, local device logs, metadata).
2. Which forensic techniques were applied (API extraction, network packet analysis, chip-off, JTAG, logical acquisition).

3. Identified security vulnerabilities and attack vectors (e.g., AvA attack, malicious skills).

4. Limitations of prior work, such as reliance on credentials, encrypted artifacts, or small sample sizes.

The findings were organized into thematic categories: cloud forensics, companion app analysis, device-level forensics, attack surfaces, and acquisition challenges, allowing structured comparison across studies. This method enabled identification of patterns, gaps, and trends, which informed the Results & Discussion chapter.

4. Results & Discussion

4.1 Comparative Summary of Previous Literature

Table 4.1: Summary of Key Literature (2020–2025)

Reference	Focus Area	Contribution	Limitation
Esposito, Sgandurra & Bella (2020)[5]	Smart-speaker self-issue vulnerability	Demonstrated the AvA attack: Amazon Echo devices can self-issue voice commands via audio files (Bluetooth or radio-based), enabling arbitrary actions (e.g. smart-home control, eavesdropping); identified two further vulnerabilities Full Volume (boosts command recognition) and Break Tag Chain (extends malicious skill runtime).	Small sample size, may vary with device models or environments, FM radio attack only works in the USA, countermeasures not tested.
Handgar et al. (2023)[7]	Security & privacy of the Amazon Alexa ecosystem	Provides a comprehensive review of known Alexa attack techniques, analyzes the security risks of third-party skills such as skill squatting and voice masquerading, and outlines mitigation strategies to strengthen the Alexa ecosystem.	No new experimental testing of the attacks.
Li et al. (2021)[14]	Alexa attack surfaces	Provides a comprehensive survey identifying six attack surfaces in the Alexa ecosystem and summarizes known attacks affecting voice capture, transmission, ASR/NLU, skills, Lambda functions, and S3 storage.	Survey-based only; no experimental validation, and many attacks depend on earlier Alexa versions.

Reference	Focus Area	Contribution	Limitation
Oladimeji & Zhou (2022)[17]	Alexa forensic storage and acquisition methods	Identifies where Alexa stores data at the software and cloud levels: iOS stores user identity, cookies, and databases in the `com.amazon.echo` directory, while Android stores limited metadata. Shows that most interactions are stored in AWS as JSON logs and .wav recordings, and demonstrates evidence acquisition using Cellebrite UFED and the Alexa Cloud Extractor tool.	Android yields limited data; some iOS artifacts are encrypted; cloud acquisition requires credentials or cookies; cloud extractor relies on unofficial APIs.
Youn, Lim, Seo, Chung & Lee (2021)[19]	IoT forensics on smart-display devices (Amazon Echo Show)	Provides the first detailed forensic analysis of the Echo Show 2nd Gen, extracting evidence from hardware, smartphone apps, and the Alexa cloud. Identifies key user artifacts and proposes a unified forensic framework for correlating multi-source traces.	Hardware acquisition is destructive and not always practical. Cloud access requires valid credentials, and artifacts may be incomplete or device-specific, limiting generalizability.
Aljneibi et al. (2025)[1]	Smart home forensic privacy + Alexa data retention analysis	Carries out forensic tests on several Alexa generations, showing that the devices sometimes record unintended audio, store local and cloud data (logs and .wav files), and keep deleted user commands for a long time. Suggests a practical forensic readiness model and points out privacy issues in Alexa smart homes.	Experiments were limited to a small set of devices; cloud access required valid credentials; unintended recordings may not be fully representative; reliance on tools such as FTK and Autopsy restricts visibility into encrypted artifacts.

4.2 Summary of Findings

This section summarizes the key findings extracted from the reviewed studies presented in Table 4.1. The objective is to identify common patterns, recurring results, and trends

across research on Amazon Alexa security, privacy, and forensic investigations.

Overall, the literature consistently highlights that cloud storage serves as the primary repository for Alexa-related forensic artifacts, including interaction logs, voice recordings, device metadata, and configuration data [17, 19, 1]. Companion mobile applications, particularly on Android, provide additional forensic evidence, such as cached logs, authentication tokens, images, and databases [17, 1]. However, iOS devices often store fewer accessible artifacts and some data may be encrypted, limiting immediate visibility [17].

Local device storage on Echo devices provides limited but potentially useful evidence, particularly when combined with hardware-level acquisition methods such as JTAG, chip-off, or eMMC memory dumps [19]. Such methods allow investigators to retrieve firmware, system logs, encryption keys, and authentication tokens, complementing mobile and cloud sources [19]. Several studies also emphasize the persistence of deleted commands or unintended audio recordings, which can remain recoverable for several weeks, highlighting gaps in data deletion and privacy mechanisms [5, 1].

Research on security vulnerabilities, including Alexa vs Alexa (AvA) attacks and third-party skill exploits, demonstrates that spoofed voice commands and misconfigured skills can compromise both security and the integrity of forensic evidence [5, 7, 14, 1]. Limitations identified across studies include small experimental sample sizes, reliance on valid credentials for cloud access, dependence on unofficial APIs, and partial encryption of some artifacts [5, 17, 19, 1].

In summary, the combined use of cloud acquisition, mobile device extraction, and low-level hardware forensic methods represents the most comprehensive approach for reconstructing user activities, device interactions, and environmental events within the Alexa ecosystem [17, 19, 1]. These findings reinforce the importance of multi-layered forensic strategies to ensure reliable and complete evidence collection.

4.3 Critical Analysis and Identified Gaps

Despite the availability of multiple sources of evidence within the Alexa ecosystem, the system still suffers from significant gaps that limit the effectiveness of forensic investigations. Alexa relies heavily on the cloud, which restricts local device analysis, particularly due to the absence of complete voice recordings on local storage [15, 17]. Accessing cloud data often requires legal orders or valid credentials, which can delay or restrict data retrieval. A more pronounced gap exists in third-party skills, where storage policies vary and oversight is not stringent, allowing attacks such as Voice Masquerading and Skill Squatting, which impersonate legitimate skills while collecting sensitive data or interacting without the user's knowledge [7, 14]. Additionally, new Alexa devices are registered with the FCC under LLC entities not directly associated with Amazon, reducing device transparency and delaying researchers' access to technical information necessary for

developing accurate and effective investigative [15].

4.4 Implications and Limitations

The findings indicate that the Alexa ecosystem carries serious security and forensic risks, particularly when considering third-party skill attacks and the Alexa vs Alexa (AvA) vulnerability. In the AvA scenario, sensitive data and user conversations handled by a malicious skill may be diverted to the attacker's server instead of Amazon's official cloud, resulting in a complete absence of traces within Amazon's logs and complicating forensic investigation [5, 7]. Moreover, malicious skills can mimic legitimate Alexa responses and operate silently, intercepting information without leaving visible evidence in standard logs [7].

These risks are aggravated by limited user awareness and weak default protection; the lack of strong voice-authentication enables external devices or prerecorded audio to issue commands without a human user present [20]. Therefore, users must adopt stricter security practices such as disabling unused features, regularly reviewing command history, and restricting permissions granted to skills.

On the corporate side, Amazon has the responsibility to enhance vetting mechanisms and restrict skills that rely on voice spoofing, and improve device transparency. This is especially important given that some Alexa devices are registered under unrelated LLC entities, complicating forensic readiness and hindering the development of investigation tools [15]. The involvement of third-party servers in data processing further raises legal and technical challenges regarding evidence integrity and chain of custody [15].

Addressing these issues requires coordinated efforts: users must raise their security awareness, manufacturers must strengthen platform protections, and data flows involving third parties must be restricted to ensure robust protection and forensic reliability within the Alexa ecosystem.

4.5 Recommendations for Future Work

Based on the identified gaps and limitations in the existing literature, several key directions are proposed to guide future research on Alexa forensics and security:

1. Conduct Experimental Forensic Analysis on Newer Echo Devices
   Most current studies focus on older Echo models. Future research should test newer devices to determine whether Amazon has modified data storage, logging behavior, or security protections. This will help validate whether existing forensic techniques remain effective.

2. Develop a Standardized Alexa Forensic Framework
   There is no unified methodology that combines cloud acquisition, mobile app analysis, and device-level forensics. Creating a comprehensive forensic framework would improve consistency, accuracy, and reliability in Alexa investigations.

3. Evaluate the Impact of Voice Spoofing Attacks on Evidence Integrity
   Attacks such as Alexa vs Alexa raise serious concerns about the authenticity of voice-based logs. Future work should investigate how voice spoofing affects the reliability and admissibility of Alexa evidence, and propose verification mechanisms.

5. Conclusion

In this paper we examined how Amazon Alexa stores user interactions across cloud services, companion applications, and local device memory. The findings show that the cloud provides the most complete forensic record, including voice logs, transcripts, and historical user activity. The companion app adds useful artifacts such as cached responses and authentication metadata, while Echo devices offer limited but meaningful local traces like system logs and media metadata. The analysis also identified several challenges, including encrypted storage, restricted local data, reliance on third-party skills, and vulnerabilities such as the AvA attack. Many forensic acquisition methods require privileged cloud access or hardware-level extraction, which may not always be feasible. Overall, the study demonstrates that reliable reconstruction of user behavior requires a multi-layered forensic approach that integrates cloud data, app artifacts, and device-level evidence. This research contributes to IoT forensics by clarifying where Alexa evidence resides, how it can be retrieved, and what limitations investigators should anticipate when analyzing smart speaker environments.

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